AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM352
Web, Mobile and Cloud Technologies

Final Examination
Fall Semester 2024/2025

Date:16/Jan./2025

Number of Exam Pages: 04 (including this cover sheet)	**Time Allowed: 3 Hrs**

Instructions:

- Total Marks: **100**
- This exam consists of **03** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1 (20 Marks)

Mark the following statements True(T) or False(F). Copy the answers in your answer booklet [2 Marks each]

Statement	T	F
1. To verify a digital signature, we need the Sender's Public key.	◯	◯
2. SOAP is a W3C recommendation that uses JSON as its message format.	◯	◯
3. User agent string is sent in page requests for performing browser sniffing.	◯	◯
4. Diffie–Hellman algorithm is used for Asymmetric key encryption	◯	◯
5. @FormParam annotation is suitable to extract the parameters from the following URI: www.example.com/products?from=100&to=200&orderBy =name	◯	◯
6. Horizon component serves as a dashboard for users to manage OpenStack compute, storage and networking services.	◯	◯
7. TCP Port number distinguish between different applications running on any one computer in a network.	◯	◯
8. In PaaS, The tenant cannot deploy applications onto the cloud infrastructure that are created using programming languages, libraries, services, and tools supported by the provider	◯	◯
9. When using a container, the contained application communicates directly with hardware.	◯	◯
10. Scaling out occurs when going from two load-balanced web server instances to three instances.	◯	◯

PART 2 (30 Marks)

Answer the following question.

1. You are given the following table that represents cars and its model.

cars	Nissan	model	Sentra	Maxima	Skyline	Ford	model	Taurus	Escort
		doors	4	4	2		doors	4	4

Complete the XML file by adding the missing elements and attributes which represent above table. [10 Marks]

```
<?xml version="1.0" encoding="UTF-8" ?>
<root>
  <cars>
    <Nissan>
  <! Complete the missing elements>
  </cars>
</root>
```


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2. Demonstrate the steps that should occur at sender and receiver to preserve the confidentiality of a message sent from Omar to Ahmed using the symmetric key method. [5 marks]

3. Describe the three solutions from which mobile web designers can choose to address the screen size issue. [5 marks]

4. Compare between containers and virtual machines with the help of diagrams. [10 marks]

PART 3 (50 Marks)

Read the scenario and solve the given four tasks.

Scenario:

A ride-hailing web application like OLA, to be deployed on a cloud, contains a cost calculator service to be used to calculate the fare required for a trip given cost per meter, ride distance, rider ID. The calculation is performed according to the following formula

Trip Fare =costpermeter x ridedistance x (1- discount percentage)
equation.1

The amount of discount percentage for each rider is stored in a json file named riders.json.

The service should accept cost per meter, ride distance, rider ID and should return the result as HTML

1) Implement the service as RESTFUL service with the required annotations and read the discount percentage given for a rider from a JSON file called "riders.json" using the classes Json, JsonObject, and JsonArray. The file contains an array of riders with each array element contains riders ID and the discount percentage. [25 Marks]
2) Design the HTML page and write the HTML code that would consume or call the service. [15 Marks]
3) Specify the parameter type you have used in your implementation and the HTTP method used to call the service. [4 Marks]
4) In case that the web application should be available to mobile users, compare the development of the application as web, native, or hybrid application regarding the development skills required. [6 Marks]

Some Json class methods that could be used:

- static JsonArrayBuilder createArrayBuilder()//Creates a JSON array builder
- static JsonObjectBuilder createObjectBuilder()//Creates a JSON object builder
- static JsonReader createReader(InputStream in)//Creates a JSON reader from a byte stream.
- static JsonReader createReader(Reader reader) //Creates a JSON reader from a character stream.

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- static JsonWriter createWriter(OutputStream out) //Creates a JSON writer to write a JSON object or array structure to the specified byte stream.
- static JsonWriter createWriter(Writer writer) //Creates a JSON writer to write a JSON object or array structure to the specified character stream.

Some JsonObject methods that could be used:

- boolean getBoolean(String name) //Returns the boolean value of the associated mapping for the specified name.
- int getInt(String name)//returns int value to which the specified name is mapped.
- JsonArray getJsonArray(String name)//Returns the array value to which the specified name is mapped.
- JsonObject getJsonObject(String name)//Returns the object value to which the specified name is mapped.
- String getString(String name)//Returns the string value to which the specified name is mapped

Some JsonArray methods that could be used

- int getInt(int index) //A convenience method for getJsonNumber(index).intValue().
- JsonArray getJsonArray(int index) //Returns the array value at the specified position in this array.
- JsonObject getJsonObject(int index) //Returns the object value at the specified position in this array.
- String getString(int index) //Returns the String value at the specified position in this array.

Some JsonReader methods that could be used

- void close() //Closes this reader and frees any resources associated with the reader.
- JsonArray readArray() //Returns a JSON array that is represented in the input source.
- JsonObject readObject() //Returns a JSON object that is represented in the input source.

Some JsonWriter methods that could be used

- void close() //Closes this JSON writer and frees any resources associated with the writer.
- void writeArray(JsonArray array) //Writes the specified JSON array to the output source.
- void writeObject(JsonObject object) //Writes the specified JSON object to the output source.

End of questions

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